REVA UNIVERSITY

School of Computer Science & Engineering

AI Application Development – IA1

SkillProof

Developer Credibility Verification Platform

Turning GitHub Repositories Into Auditable Hiring Evidence

www.reva.edu.in

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Team Members & Roles

Team member name	Role
Harsha B U - R23EF089	TEAM LEAD
Gowtham M S - R23EF085	DEVELOPMENT LEAD
Arjun reddy , Arnav sharma - R23EF030 , R23EF031	CUSTOMER LEAD | TEST / CUSTOMER
Harsha B U, Ganesh Hugar – R23EF089,R23EF080	DEVELOPER
Gowtham M S , Dattatri – R23EF085,R23EF064	DEVOPS AND DOCUMENTATION
Amarendra Murthy – R23EF019	TEST TEAM

Problem Statement

GitHub profiles do not prove that candidates actually understand the code they claim to have built — especially after AI tools made generating impressive-looking repositories trivially easy.

Portfolio Inflation

Candidates present polished GitHub repos that were AI-generated or copied, without genuine understanding of the codebase.

AI-Assisted Code

LLMs make it easy to produce well-structured repositories, making traditional portfolio review unreliable as a signal.

Expensive False Positives

Senior engineers waste interview bandwidth on candidates whose claims evaporate at the first deep technical question.

Fragmented Process

Assessment, proof-of-work, and recruiter review happen across disconnected tools — no single evidence trail exists.

We did not choose a convenience problem — we chose a trust problem where wrong hiring decisions cost companies money and disadvantage genuine candidates.

User / Customer Understanding

Recruiters & Hiring Managers

- Need fast, reliable candidate filters before deep rounds
- Frustrated by inflated GitHub portfolios
- Want auditable evidence — not just a score
- Spend excessive time on candidates with exaggerated skills

Candidates (Developers)

- Genuine developers struggle to differentiate from AI-assisted submissions
- No credible way to prove authorship of their own code
- Portfolio-only screening feels superficial and unfair
- Want a platform that validates real understanding publicly

Interview Insights

"I can't tell if they actually wrote the code or just used Copilot" – Recruiter, SaaS startup

"My GitHub profile looks impressive but I get rejected in first technical rounds" – Junior Dev

"We need something between resume screening and live coding" – Engineering Manager

Ideal Customer Profile for 'SkillProof'

Ideal Customer Profile (ICP)

Startups

Demographic: 10–200 employees, tech-first, scaling engineering team

Psychographic: Speed and quality of hire matter equally; want signal without bureaucracy

Key Pain: Screened candidates arrive unprepared; senior devs spend days interviewing

Mid-size Tech Firms

Demographic: Remote-first, hiring 5–30 developers/quarter

Psychographic: Risk-averse on remote hires; want documented hiring trails

Key Pain: Cannot verify GitHub project authenticity at scale; ATS gives no depth

Training Institutes

Demographic: Bootcamps, universities with placement cells

Psychographic: Want to prove student employability; need objective project proof

Key Pain: Faculty cannot individually verify capstone project depth

Customer Persona (CP)

AK

Arjun Kulkarni

Tech Recruiter – Bangalore

Age: 29

Experience: 5 yrs hiring

Team Size: 8-person HR tech

Hires/Quarter: 15–25 devs

Goals

✓ Screen 50+ candidates/week efficiently

✓ Build evidence before spending engineer time

✓ Reduce bad hires from inflated portfolios

Frustrations

✕ Candidates can't explain their own GitHub code

✕ AI assistance makes portfolios look deceptively strong

✕ No single tool ties portfolio → verification → interview

Product USP – Why SkillProof?

1 Repo-Grounded Questions

Questions are generated from the candidate's actual code — not a generic question bank. Every question references real file names, methods, and architectural decisions from their repository.

2 Cryptographic Evidence Trail

Answers, scores, behavior signals, and integrity metrics are sealed in an HMAC-SHA256 signed badge. Evidence is tamper-proof and auditable by recruiters at any time after the session.

3 Live One-to-One Verification

For borderline cases, recruiters launch a controlled live session where questions are revealed one by one — preventing pre-generated AI answer abuse and forcing real-time reasoning.

4 Execution-Based Proof Layer

Docker-sandboxed hidden-test challenges validate that candidates can write working code — providing objective pass/fail evidence that text-answer AI cannot fake.

System Architecture

Frontend

Next.js + TypeScript

- GitHub OAuth login
- Repo selector UI
- Answer submission form
- Telemetry capture
- Recruiter dashboard

Backend Core

Spring Boot (Java)

- Auth & session mgmt
- File ingestion pipeline
- Scoring engine
- Badge + HMAC signer
- Recruiter & live APIs

Backend AI

FastAPI (Python)

- Question generation
- Answer evaluation
- Rubric scoring
- Gibberish detection
- Reference answer gen

Docker Execution Service

Sandbox: no network · memory cap · CPU cap · read-only FS · tmpfs temp storage

GitHub REST API

Repo tree · file contents · metadata · language detection (no local clone)

End-to-End Verification Pipeline

1	2	3	4	5	6	7	8
GitHub OAuth	**Repo Select & Ingest**	**AI Question Gen**	**Candidate Answers**	**AI Evaluation**	**Score + Badge**	**Recruiter Review**	**Live / Challenge**
Candidate authenticates; repo list fetched	File tree → filter → rank → extract code summary	5–7 repo-grounded questions with difficulty mix	Timed responses; telemetry captured per question	Accuracy(40%) + Depth(30%) + Specificity(30%)	Integrity-adjusted score; HMAC-signed badge issued	Evidence panel: questions, answers, signals, refs	One-to-one reveal mode or Docker hidden-test

Anti-Cheat Layers: Specificity Guards | Identifier Count Caps | Gibberish Detection | Tab Switch & Paste Telemetry | Live Reveal Mode | Docker Hidden Tests

Scoring Formula

questionScore = (Accuracy × 4 + Depth × 3 + Specificity × 3) / 10

integrityAdjustedScore = max(0, technicalScore – penalties)

Penalties: tabSwitches + pasteCount + copyEvents + anomalousSpeed

Badge Contents

- ✓ Score (technical + integrity-adjusted)
- ✓ Skill dimension breakdown (5 axes)
- ✓ Per-question evidence + timing
- ✓ Behavior telemetry (tab/paste/copy)
- ✓ HMAC-SHA256 signed token

Use Cases & High-Level Requirements

UC-1: Async Verification

Actor: Candidate

Connects GitHub, selects repo, answers AI-generated questions, receives integrity-aware badge.

IN: GitHub repo URL

OUT: Signed badge with score

UC-2: Recruiter Evidence Review

Actor: Recruiter

Views candidate list, inspects question-level evidence, reveals full answers, compares with AI reference.

IN: Badge token

OUT: Evidence panel + reference answers

UC-3: Live One-to-One Session

Actor: Recruiter + Candidate

Recruiter reveals questions one by one in real time, candidate answers only the active question. Score computed live.

IN: Session invite code

OUT: Live score + evidence log

UC-4: Execution Challenge

Actor: Recruiter + Candidate

Recruiter creates hidden-test challenge; candidate submits code; Docker sandbox runs tests and returns pass/fail evidence.

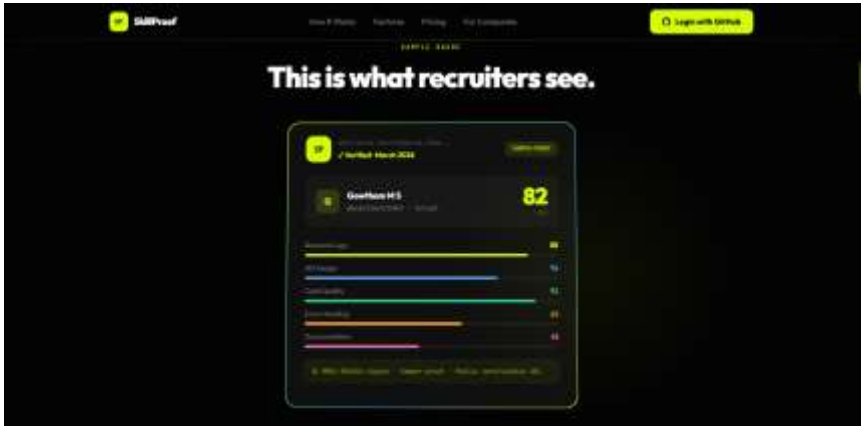
IN: Code submission

OUT: Test pass/fail report

Proof of Concept (PoC) / Platform Demo

Feature / Component	Status
GitHub OAuth + Repo Selection	Complete
AI Question Generation (code-grounded)	Complete
AI Answer Evaluation + Rubric Scoring	Complete
Integrity Telemetry (tab / paste / copy)	Complete
HMAC Badge Issuance	Complete
Recruiter Dashboard + Evidence Panel	Complete
Live One-to-One Session Mode	Complete
Docker Challenge Execution	Configurable
ATS Integration / Export Report	Roadmap
Enterprise RBAC + Org Workspaces	Roadmap

SP
SkillProof
How It Works
Features
Pricing
For Companies
Login with GitHub
GitHub shows what you built. We prove you understand it.
AI questions grounded in your actual code.
Verified badges recruiters trust. No generic tests.
Verify My Portfolio Free
See How It Works ↓


SkillProof
This is what recruiters see.
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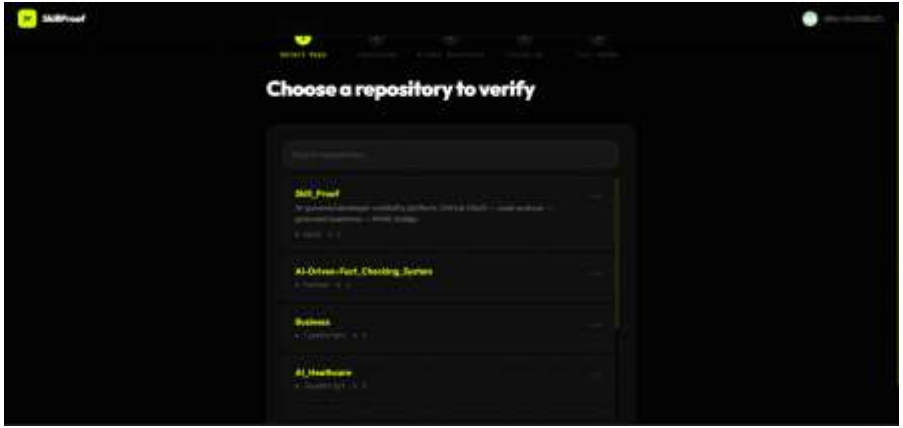
SkillProof
HMAC Badge

SkillProof
HMAC Badge


SkillProof
Choose a repository to verify

Limitations & 90-Day Improvement Roadmap

Known Limitations

- ⚠ Async mode can't prevent sidebar AI (phone) use
- ⚠ LLM scoring has inherent probabilistic variance
- ⚠ File key collision risk on same-name files
- ⚠ Two-service ops overhead (no circuit breaker yet)
- ⚠ Frontend uses localStorage (not HttpOnly cookie)
- ⚠ No formal evaluator benchmark harness yet

Mitigations & Next Steps

- → Live reveal mode forces real-time answers
- → Deterministic quality guards cap AI inflation
- → Migrate to full relative path keys in content map
- → Add circuit-breaker & retry around AI gateway
- → Migrate token to HttpOnly cookie in prod build
- → Build labeled benchmark answer set for calibration

0–30 Days	Harden evaluator guardrails · Finish manual recruiter verdict actions · Risk indicator UI
31–60 Days	Enable Docker challenges in staging · Exportable PDF evidence report
61–90 Days	ATS integration · Org workspaces + RBAC · Hiring funnel analytics dashboard